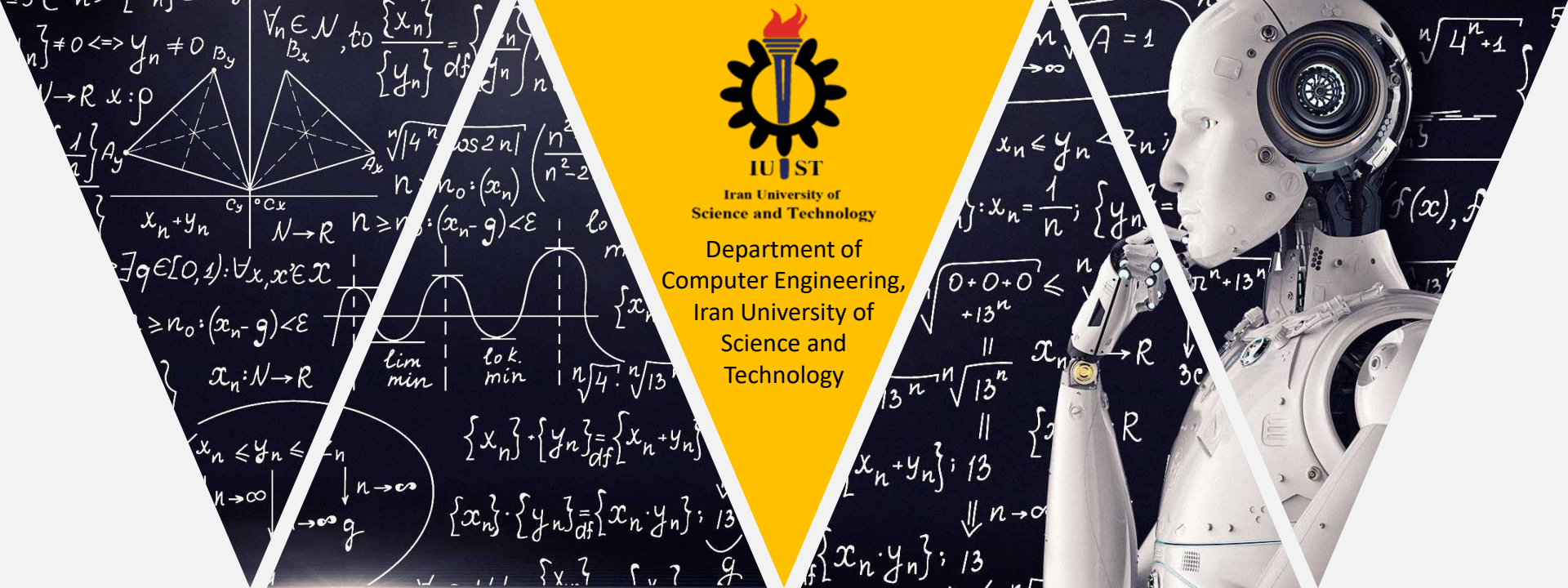




Machine Learning: Parametric and Non-Parametric Testing

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Overview

- K-Nearest Neighbors Algorithm
- Statistical t-Test
- Kolmogorov-Smirnov Test
- Conclusion



K-Nearest Neighbors Algorithm

Introduction

❖ Supervised Machine learning algorithm

- Classification
- Regression

❖ Instance-Based Learning

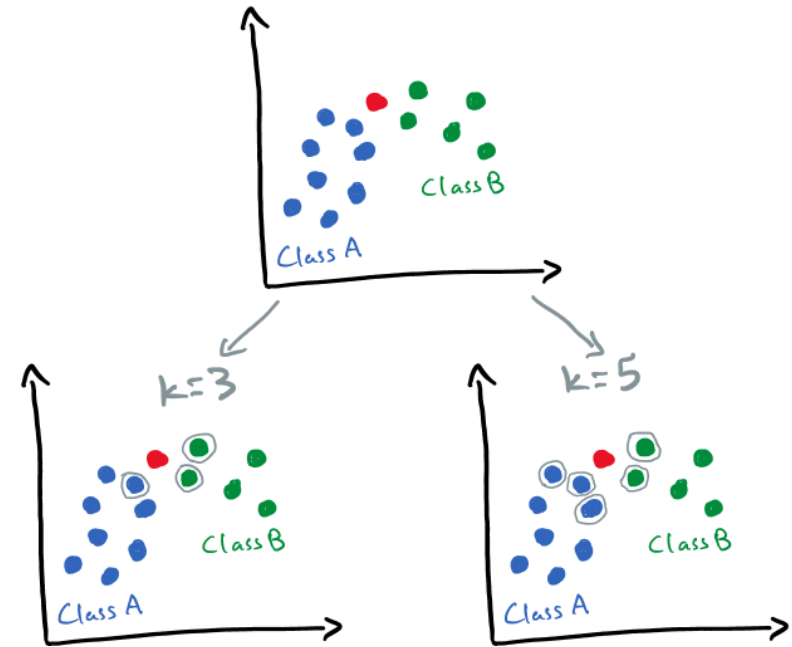
- Stores all training data
- Lazy learning: No model training phase

❖ Voting System

- Majority vote (classification)
- Average (regression)

❖ Hyper-parameters

- Number of neighbors (k), distance metric and weighting scheme



Ref: ealizadeh.com/blog/knn-and-kmeans/



K-Nearest Neighbors Algorithm

Important Points

❖ Non-parametric

- No assumption about data distribution

Pros	Cons
Simple and easy to implement	Computationally expensive for large datasets
Performs well with low-dimensional data	High memory usage due to storing all data
Adaptable to different applications	Sensitive to noise and outliers
Supports various distance metrics	Feature scaling is required for proper distance calculation



K-Nearest Neighbors Algorithm

Important Points

❖ Density Estimation with KNN

- Estimate the probability density of data points in a dataset.

$$\tilde{P}_{KNN}(x) = \frac{K}{N} * \frac{1}{V_d * R_K^d(x)}$$

$$V_d = \frac{\pi^{\frac{d}{2}}}{\Gamma(\frac{d}{2} + 1)}$$

$$\Gamma(n) = (n - 1)!$$



Statistical t-Test

Introduction

❖ Statistical method used to compare means under different conditions

- significant
- randomness and sampling fluctuations

❖ Parametric Test

- require specific assumptions about the distribution of data

Normal Distribution

Interval or Ratio Data

Homogeneity of
Variances

- In the **independent t-Test**, the variances of the two groups should be equal
Otherwise using **Welch's t-Test** .



Statistical t-Test

states

❖ One-Sample t-Test

- Comparison of the sample mean with a known fixed value.

❖ Two Independent Sample t-Test

- Comparison of the means of two independent populations with equal variance.
- **EXP**: Comparison of the average height of two groups of people.

❖ Paired Sample t-Test

- Comparison of two dependent variables with equal variance.
- Used when a quantitative variable is measured twice per observation
- **EXP**: Effect of blood pressure medication on patients before and after administration



Statistical t-Test

Formula

❖ One-Sample t-Test

- Comparison of the sample mean with a known fixed value.

$$t = \frac{\bar{X} - \mu_0}{\frac{S}{\sqrt{n}}}$$

$$s = \sqrt{\frac{\sum_{i=1}^n (x_i - \bar{X})^2}{n - 1}}$$

$$df = n - 1$$



Statistical t-Test

Example Code

❖ One-Sample t-Test

- Comparison of the sample mean with a known fixed value.

```
import numpy as np
from scipy import stats

data = np.array([13, 15, 9, 20, 16, 18, 15, 10, 11, 13]) # Sample data
t_stat, p_value = stats.ttest_1samp(data, 13) # Compare with  $\mu=13$ 

print(f"One-Sample T-Test: t={t_stat:.4f}, p={p_value:.4f}")
```

Statistical t-Test

Formula

❖ Two Independent Sample t-Test

- Comparison of the means of two independent populations with equal variance.
- **EXP**: Comparison of the average height of two groups of people.

$$t = \frac{\overline{x_A} - \overline{x_B}}{\sqrt{\frac{s^2}{n_A} + \frac{s^2}{n_B}}}, \quad df = n_A + n_B - 2$$

$$s^2 = \sqrt{\frac{\sum_{i=1}^{n_A} (x_i - \overline{x_A})^2 + \sum_{i=1}^{n_B} (x_i - \overline{x_B})^2}{n_A + n_B - 2}}$$



Statistical t-Test

Example Code

❖ Two Independent Sample t-Test

- Comparison of the means of two independent populations with equal variance.

```
import numpy as np
from scipy import stats

group1 = np.array([10, 12, 14, 16, 18, 20])
group2 = np.array([11, 13, 15, 17, 19, 21])

t_stat, p_value = stats.ttest_ind(group1, group2, equal_var=False) # Assuming unequal variance
print(f"Independent T-Test: t={t_stat:.4f}, p={p_value:.4f}")
```



Statistical t-Test

Formula

❖ Paired Sample t-Test

- Comparison of two dependent variables with equal variance
- Used when a quantitative variable is measured twice per observation
- **EXP**: Effect of blood pressure medication on patients before and after administration

$$d_i = x_{after,i} - x_{before,i}$$

$$\bar{d} = \frac{1}{n} \sum_{i=1}^n d_i$$

$$t = \frac{\bar{d}}{\frac{S}{\sqrt{n}}}$$

$$df = n - 1$$

Statistical t-Test

Example Code

❖ Paired Sample t-Test

- Comparison of two dependent variables with equal variance
- Used when a quantitative variable is measured twice per observation

```
import numpy as np
from scipy import stats

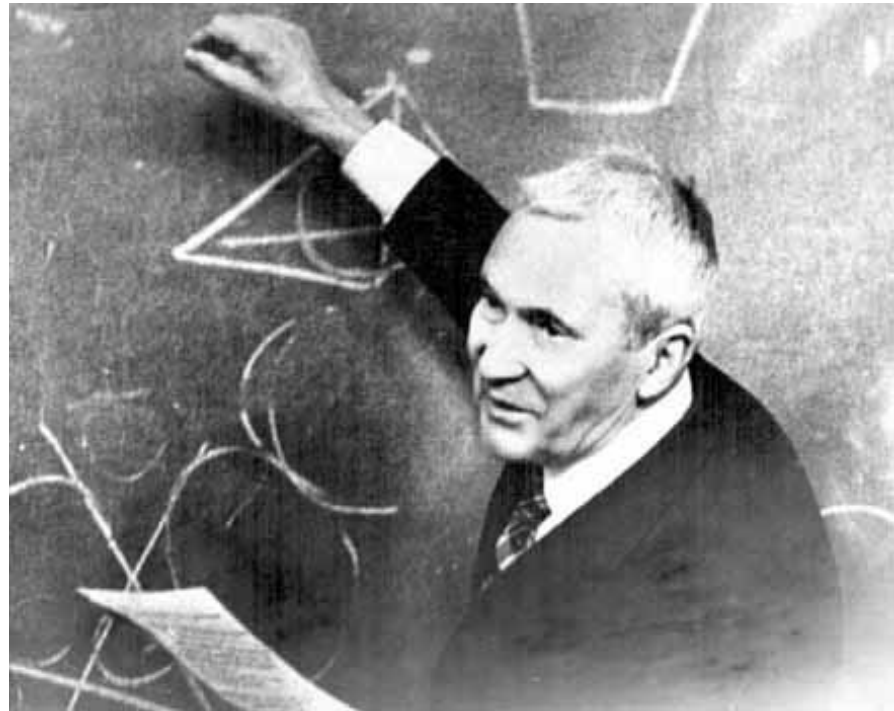
before = np.array([80, 82, 85, 78, 90, 88])
after  = np.array([85, 84, 87, 80, 93, 91])

t_stat, p_value = stats.ttest_rel(before, after)
print(f"Paired T-Test: t={t_stat:.4f}, p={p_value:.4f}")
```

Kolmogorov-Smirnov Test

Introduction

- ❖ Two 21th-century Russian mathematicians
- ❖ Non-Parametric Test



Kolmogorov-Smirnov Test

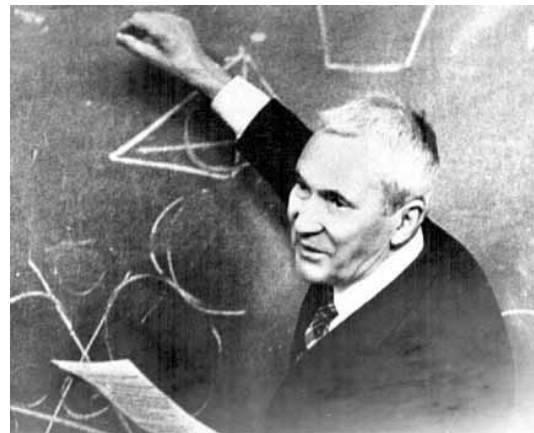
states

❖ One-sample K-S test

Examines whether a sample follows a specific distribution or not.

❖ Two-sample K-S test

Examines whether two independent samples come from the same distribution or not.





Kolmogorov-Smirnov Test

states

❖ One-sample K-S test

- Examines whether a sample follows a specific distribution or not
- step one (ECDF) : Calculate Empirical Cumulative Distribution Function

$$F_n(x) = \frac{1}{n} \sum_{i=1}^n 1, \text{ if } X_i \leq x$$

- step two : calculate Theoretical Cumulative Distribution Function
- step three : test statistic

$$D_n = \sup_x |F_n(x) - F(x)|$$

Kolmogorov-Smirnov Test

Example Code

❖ One-sample K-S test

- Examines whether a sample follows a specific distribution or not.

```
import numpy as np
from scipy.stats import ks_1samp, norm

# Generate a sample from a normal distribution
np.random.seed(42)
sample = np.random.normal(loc=0, scale=1, size=1000)

# Perform the one-sample KS test
ks_stat, p_value = ks_1samp(sample, norm.cdf)

print("One-sample KS test:")
print(f"KS Statistic: {ks_stat:.4f}, P-value: {p_value:.4f}")

# Interpret the result
if p_value > 0.05:
    print("Fail to reject H0: Sample follows the normal distribution.")
else:
    print("Reject H0: Sample does not follow the normal distribution.")
```



Kolmogorov-Smirnov Test

states

❖ Two-sample K-S test

- Examines whether two independent samples come from the same distribution or not
- step one (ECDF) : Calculate Empirical Cumulative Distribution Function

$$F_m(x) = \frac{1}{m} \sum_{i=1}^m 1, \text{ if } X_i \leq x \quad G_n(x) = \frac{1}{n} \sum_{i=1}^n 1, \text{ if } Y_i \leq x$$

- step two : test statistic

$$D_{m,n} = \sup_x |F_m(x) - G_n(x)|$$

Kolmogorov-Smirnov Test

Example Code

❖ Two-sample K-S test

- Examines whether two independent samples come from the same distribution or not

```
import numpy as np
from scipy.stats import ks_2samp

# Generate two samples from different distributions
np.random.seed(42)
sample1 = np.random.normal(loc=0, scale=1, size=1000) # Normal distribution
sample2 = np.random.uniform(low=-2, high=2, size=1000) # Uniform distribution

# Perform the two-sample KS test
ks_stat, p_value = ks_2samp(sample1, sample2)

print("Two-sample KS test:")
print(f"KS Statistic: {ks_stat:.4f}, P-value: {p_value:.4f}")

# Interpret the result
if p_value > 0.05:
    print("Fail to reject H0: The two samples may come from the same distribution.")
else:
    print("Reject H0: The two samples come from different distributions.")
```



Conclusion

❖ Statistical t-Test

- Sensitive to Outliers
- The data must follow a normal distribution
- The t-Test is only applicable for comparing two means.
 - should use ANOVA to compare more than two groups

❖ Statistical Kolmogorov-Smirnov Test

- Non-parametric and applicable to any type of distribution
- If the data is discrete, the KS test may not perform accurately.



THANK YOU

